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Nambur, Pedakakani (M), Guntur (Dt) - 522508
Department of CSE - Artificial Intelligence & Machine Learning

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented ☐ Application oriented ☐ Product oriented ☐ Other (please specify) Specify if others:	Batch ID:	AIML A14

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Title of the Project: Attack-Resistant Swarm Learning for Medical Diagnosis

Abstract

Swarm Learning is a method that allows multiple organizations to work together to train an Artificial Intelligence model without sharing their private data. It is useful in areas where data is sensitive and cannot be directly shared. However, Swarm Learning has a major problem: a malicious participant can send incorrect or harmful model updates to affect the performance of the overall model. This is known as a poisoning attack. Another challenge is that different organizations may have different types and amounts of data. Because of this, their model updates can naturally be different even when they are honest. A system that checks only model similarity may therefore wrongly identify an honest participant as malicious.To solve this problem, we propose an adaptive multi-factor trust mechanism for Swarm Learning. In this approach, each participant trains a local model using its own data and shares only the model update instead of the original data. The proposed method checks each update using multiple factors, including cosine similarity, data distribution, local validation performance, update magnitude, and previous participation history. Based on these factors, a trust score is calculated for each participant. Trusted updates are given more importance, while suspicious updates are given less importance during the final model combination.The proposed system aims to improve the security and reliability of Swarm Learning while reducing the chance of rejecting genuine participants. The system will be tested under normal conditions, different data distributions, and poisoning attacks. The performance will be evaluated using F1 Score, Attack Success Rate, and False Positive Rate.

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References (Category specific)

[1] S. Warnat-Herresthal *et al.*, "Swarm learning for decentralized and confidential clinical machine learning," *Nature*, vol. 594, pp. 265–270, 2021, doi: 10.1038/s41586-021-03583-3.

[2] X. Cao, M. Fang, J. Liu, and N. Z. Gong, "FLTrust: Byzantine-robust federated learning via trust bootstrapping," in *Proc. Network and Distributed System Security Symposium (NDSS)*, 2021.

[3] C. Fung, C. J. M. Yoon, and I. Beschastnikh, "Mitigating Sybils in federated learning poisoning," *arXiv preprint arXiv:1808.04866*, 2018.

[4] G. Xia, J. Chen, C. Yu, and J. Ma, "Poisoning attacks in federated learning: A survey," *IEEE Access*, vol. 11, pp. 10708–10722, 2023, doi: 10.1109/ACCESS.2023.3238823.

[5] V. Tolpegin, S. Truex, M. E. Gursoy, and L. Liu, "Data poisoning attacks against federated learning systems," in *Proc. 25th European Symposium on Research in Computer Security (ESORICS)*, 2020.

Project Team Members

S. No.	Regd. No.	Name of the Student	Signature
1	23BQ1A6129	D. Bala Thripura Sundari	
2	23BQ1A6136	G. Durga Harsha Vardhan	
3	23BQ1A6123	Ch. Teja Sri	
4	24BQ5A6125	G. Rakesh	

Project Guide

Name: Signature:

MAJOR PROJECT REGISTRATION FORM

Date:

INSTRUCTIONS

(No need to submit this page so you don't have to take the print)

Project category

Project categories were mentioned only for the sake of documentation. All of you must chose '**Research oriented**' projects only. Very few projects would be accepted if the project statements are chosen from Smart India Hackathon (SIH) or from any other reputed platform for that matter. In such cases, you have to write the problem ID mentioned in those platforms as it is in the '**Project category**' section. Also provide the link to the problem statement in the '**References**' section.

To write Abstract

(Use the following as a mental checklist)

a) **Background & Problem Statement:** What is the exact real-world problem your project addresses? State it precisely and narrowly not "we are improving healthcare" but "patients in rural areas lack timely access to dermatology consultation."
b) **Target Users / Stakeholders:** Who exactly faces this problem? A defined persona, not "everyone" e.g. "first-generation college students applying for scholarships."
c) **Existing Gap & Evidence:** What do people currently do to cope with this problem, and why is it inadequate? Cite any survey, interview, article, or report that confirms the problem is real.
d) **Related Work / Competitive Landscape:** What are the existing tools/apps/products/papers that partially address this, and how will yours differ or improve on them? (Feeds directly into your References in Section 3.)
e) **Proposed Approach (overview):** One paragraph describing your approach/technique/system at a high level no implementation detail needed yet.
f) **Scope & Boundaries:** What will you solve, and explicitly what will you NOT attempt within this project's timeline?
g) **Expected Outcome / Success Criteria:** How will you know the project succeeded? State one or two measurable, testable outcomes (e.g. "90% of test users complete the task without guidance").

To write category specific references

Research oriented	Cite 1 base paper your problem statement builds on. The paper must be published within the last 3 years (2023 onward as of 2026). Add 2-4 supporting papers from the literature review if relevant.
Application oriented	Identify 2-3 existing, similar applications to use as your benchmark. For each, cite the app/website itself (as a web reference) and, if available, any article or documentation describing it.
Product oriented	Identify 2-3 existing, similar products (hardware/software) to use as your benchmark. Cite the manufacturer's page, datasheet, or a credible review/comparison article for each.

Project Coordinator HOD-CSE (AI)